

TEAM OVERVIEW

ELASTIQ retail pricing decision studio

A visual, plain-language guide to how the application turns retail history and market evidence into controlled next-period pricing recommendations.

AUDIENCE	PRIMARY DECISION	OPERATING MODE
Commercial and technical teams	Raise, hold, or lower price	Reviewed decision support

THE APPLICATION IN ONE PAGE

What ELASTIQ does

ELASTIQ recommends the most suitable next-period price for each store-product decision unit, or explicitly holds the current price when the evidence is not strong enough.

8 ANALYTICAL STAGES	6 OPTIMIZATION METHODS	50-2,000 TEST-CASE SIZE	5 VISIBLE RUN STEPS
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ELASTIQ combines five questions that are normally answered separately:

Question	How the application answers it
Can we trust the input?	Validation checks completeness, consistency, types, ranges, duplicates, and business rules.
How does price affect demand?	Descriptive and causal elasticity models estimate the demand response to a price change.
Will a promotion create incremental demand?	Promotion uplift is estimated and allowed into pricing only when readiness gates pass.
What will demand be next period?	A time-safe forecast creates the do-nothing quantity baseline.
What price is commercially feasible?	Optimization evaluates candidate prices against margin, demand, competitor, inventory, and implementation controls.

The application supports review and approval. It does not automatically publish a price.

INITIAL DATA ANALYSIS

Understand the input before running

Every decision starts with a profile of the exact rows entering the optimizer. This makes data gaps, unusual commercial structure, and evidence readiness visible before recommendations are calculated.

Input group	Example fields	What the initial analysis explains
Identity	Item, product, store, category, region	How many distinct decisions are loaded and how they are distributed.
Current economics	Current price, unit cost, competitor price	Price range, average margin, and average premium or discount to competitors.
Demand and capacity	Baseline forecast, available inventory	Total and median demand plus the share of rows under inventory pressure.
Price evidence	Elasticity, source, causal flag, confidence	Elasticity range and how much of the portfolio has decision-ready causal evidence.
Promotion / market	Uplift, readiness, market signal	Promotion readiness and the bounded market context available to candidate scenarios.

100% NUMERIC COMPLETENESS	0 INVALID ROWS	6 CATEGORY VIEWS	Live EDIT FEEDBACK
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The Data page updates the profile immediately after an edit, row addition, or CSV import. A category table then shows price, margin, forecast demand, inventory risk, causal coverage, and promotion readiness side by side.

A high completeness score is necessary but not sufficient. Causal coverage, forecast provenance, feasible margin, inventory, and promotion readiness still determine whether a row may be repriced.

FROM RAW EVIDENCE TO A DECISION

The complete journey



Every stage produces evidence or a gate. Only Optimize selects a price.

Act	Stages	What the team receives
1. Establish truth	01 Validation, 02 Exploration	A trusted analytical dataset and clear commercial context.
2. Estimate response	03 Elasticity, 04 Causal test, 05 Promotion, 06 Forecast	Causal price sensitivity, promotion readiness, and next-period baseline demand.
3. Make a decision	07 Market evidence, 08 Optimization	A bounded recommendation with expected results, constraints, provenance, and status.

Every item remains visible throughout the flow. If a required input or evidence gate fails, the final output records a hold and a machine-readable reason instead of silently dropping the item.

DECISION LOGIC

How one price is selected

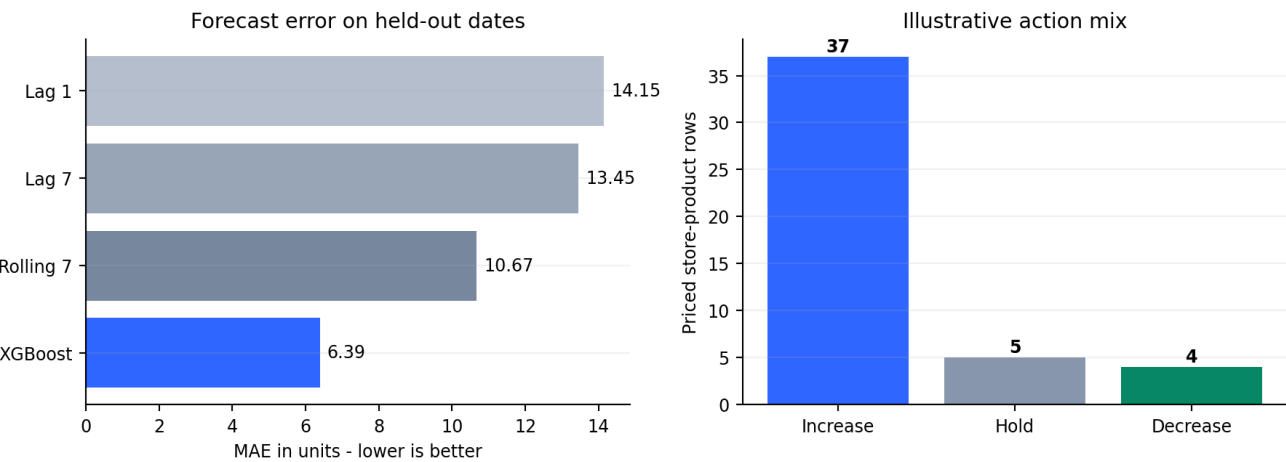


If a required gate fails, the item is held with a reason.

Step	Decision made
1. Establish the baseline	Use the forecast quantity at the current price for the next selling period.
2. Generate candidates	Create allowed prices inside configured increase and decrease limits.
3. Estimate response	Apply causal elasticity and only eligible bounded promotion and market adjustments.
4. Calculate economics	Cap realized sales at inventory and calculate comparable revenue and profit.
5. Enforce controls	Reject candidates that break margin, quantity, profit, competitor, inventory, or step rules.
6. Select and audit	Choose the best feasible candidate, apply a no-harm fallback, and recompute every control independently.

OUTPUTS AND RESPONSIBILITIES

What the team sees



Role	What they use	Typical action
Pricing / category manager	Recommended price, expected demand, profit change, binding constraints, evidence status	Approve, hold, or document an override
Revenue / finance	Portfolio and category summaries, margin and profit controls	Confirm the commercial objective and downside limits
Data science	Elasticity diagnostics, forecast metrics, promotion readiness, evidence provenance	Monitor quality, recalibrate, and investigate holds
Engineering	Pipeline status, schemas, exports, parity and regression tests	Operate releases, monitor failures, and preserve traceability
Leadership	Action mix, portfolio deltas, readiness and exception counts	Approve pilot scope and governance

Illustrative results use the packaged synthetic dataset. Live commercial performance must be measured with a controlled rollout and holdout group.

SIMPLE OPERATING WORKFLOW

Using the application

Moment	User action	Application response
Start	Run start.bat on Windows or ./start.sh on Linux/macOS.	Installs web dependencies on first use and opens the workbench.
Load	Use packaged data or import a compatible CSV.	Validates columns, values, and evidence provenance.
Configure	Choose objective, technique, price limits, margin, competitor, inventory, promotion, and implementation controls.	Shows the active decision policy before optimization.
Run	Execute the pricing analysis.	Returns item recommendations, scenario comparisons, portfolio KPIs, and control status.
Review	Inspect large moves, holds, binding constraints, and evidence quality.	Preserves reasons and provenance for every decision row.
Export	Download CSV or PDF.	Creates a review-ready decision package.

Optional: ``start.bat --refresh-data`` or ``./start.sh --refresh-data`` creates an isolated Python environment, reruns all eight analytical stages, updates the browser dataset, and then launches the application.

DEEP ANALYSIS RUNNER

Run and watch live experiments

The Deep Analysis Runner is the application's opening workflow. It executes refinement, stress re-optimization, uncertainty simulation and validation against the exact generated rows and active policy. Recommendations are not precomputed or replayed.

1. Choose the problem	2. Choose the approach	3. Watch and review
Small (50), Medium (250), Large (1,000), or custom 5-2,000 decision units. Select balanced, inventory-constrained, promotion-intensive, or volatile conditions; every dataset has a reproducible seed.	All six methods are selected by default. Choose Standard, Deep analysis, or Research workload. Deep is the default; Hybrid routes feasible items by inventory, uncertainty, promotion, and objective context.	Follow Start/profile, optimize/refine, stress test, risk simulation, validation, and Finish. Watch exact optimizer and candidate counters, runtime, events, live charts, risk percentiles, consensus and top decisions.

The workload runs in a dedicated browser worker, keeping the page responsive. Deep analysis uses three price-grid resolutions, size-adjusted shocked-market re-optimization and 1,000 Monte Carlo draws per method. Research uses four resolutions, more stress states and 3,000 draws. There is no artificial wait. A completed run retains the frozen configuration, convergence, stress distribution, risk percentiles, exact work counters, decisions and analysis report.

Deep-mode QA case	Measured compute time	Candidate evaluations
Small / 50 units / six methods	5.4 seconds	16.2 million
Medium / 250 units / six methods	9.2 seconds	28.6 million
Large / 1,000 units / six methods	26.9 seconds	62.7 million

Times were measured during release QA on the build environment and will vary by computer. Exact operation counts are reported by every live run.

Approach	Use it when	Result
Single	You need one transparent answer or want to test a chosen technique.	One measured method result.
Compare	You want a fair benchmark under the same rows and policy.	Side-by-side profit, revenue, runtime, movement, and readiness.
Hybrid	Different items may benefit from different optimization logic.	One feasible decision per item plus a method-contribution breakdown.

Run history is available for the current browser session. Any completed run can be reopened with its charts and analysis or downloaded as a full JSON record or two-page Deep Analysis Run PDF for sharing and audit.

RESPONSIBLE USE

Controls, expectations, and next steps

The application is ready for	Additional controls required for live publishing
Team demonstrations and training	Authenticated source connections and schema contracts
Reviewed scenario analysis	User roles, approval workflow, and audit retention
Synthetic and shadow-data validation	Model registry, monitoring, drift alerts, and rollback
Controlled assortment pilots	Retailer-specific causal validation and legal/policy review
CSV/PDF decision packages	Controlled price publishing integration

Recommended team practice: treat every recommendation as evidence-backed advice, preserve all holds and overrides, and measure outcomes against a predefined control group.

For implementation details, formulas, schemas, testing, and step-by-step functionality, use the companion ELASTIQ Technical System Guide.